Step 2 : Topic identification. In this step, the annotator asked to identify the topic for every cluster or sub-cluster formed. The topic must first be an event, therefore it must contain at least two components: `participant(s)` and `action`. The topic must be specific about the state or action that takes place. At the same time, the topic must be written in a way that makes it general or abstract enough to include all event sentences. Whenever possible, it is acceptable to use the most representative event sentence in a cluster as the topic.

Intermediate processing. In Steps 1 and 2, we divide the collection of clusters into 7 batches. Each of the batch contains 60 clusters and two workers are asked to annotate one same batch of clusters. This results in one cluster having two annotation results. Subsequently, an algorithm is run to automatically unify the results from two annotators. For every cluster in the original data, the algorithm starts by randomly selecting an event as a centroid. It then forms a sub-cluster around the centroid that contains all other events that are considered by both annotators to be semantically related to the centroid. The topic assigned to that sub-cluster is presented in the format `TOPIC : [Text 1] / [Text 2]` where `[Text 1]` is the topic assigned to events in this sub-cluster by the first annotator and `[Text 2]` is the topic assigned to them by the second annotator. Repeat the process with the other events until all instances are processed. Thereafter, any event that is not assigned to any cluster will exist as a stand-alone instance and temporarily be considered an outlier.

The next steps focus on resolving the disagreements from two annotation results, which includes **Topic alignment** and **Outliers processing**. We assume that a sub-cluster is properly annotated if it (1) contain at least 2 instances and (2) no annotators consider that sub-cluster to be an outlier.

Step 3: Topic alignment. Every cluster is now annotated with two topics. If both topics describe the same event, the annotator is asked to choose either or the one more representative. Otherwise, choose the one that fits most of the sentences in the cluster. If the chosen topic is already assigned to some previous cluster, merge the current cluster into that cluster. If at least one of the topics is Outliers (i.e., at least one annotator considers the sub-cluster as Outliers), temporarily view them as Outliers.

Step 4: Outliers processing. The annotator moves on to process the outliers. For any event that is assigned by only one of the previous annotators to be outliers while assigned by the other to be associated some existing sub-cluster, the annotator is asked to merge it into the assigned sub-cluster if the event can be represented by the topic of that sub-cluster; otherwise, keep it as an outlier. For any event that is agreed by both annotators to be an outlier, the current annotator is asked to re-examine it for possible assignment to any existing sub-cluster. The merging decision must be again based on the conditions described in Step 1. Any remaining stand-alone instances are discarded.

Step 5: Topic matching. This step aims to correct for potential mis-clustering from the automatic procedure. We obtain the outlier events and attempt re-categorize them into the post-annotated clustering results from all above steps. For each outlier, we present the annotators with a set of candidate clusters to which adding the outlier would not violate causal consistency. We ask them to select one cluster with whose topic the outlier is most semantically similar. The rules to determine semantic similarity of a sentence pair follows from Step 1. It is possible that there is no topic that matches the outlier. If there is any topic that is a word-by-word exact match, that topic must be selected. We also add another rule that requires the annotators to select the topic with the same level of abstraction (generality) or concreteness (specificity) as the outlier event, since there are some topics that are abstract or concrete versions of other topics. More specifically, if the outlier is concrete but the concrete topic is not presented for selection, select the abstract topic. If the outlier is abstract but the abstract topic is not presented for selection, the concrete topic must <u>not</u> be selected.

B.2 Causal Relations Discovery

The annotator is tasked with evaluating candidate pairs of clusters to determine whether a cause-and-effect relationship exists between them, based on their respective topics. Since each cluster's topic represents an event abstraction, and in essence, an event itself, the decision on causal relation hinges on whether the two topics describe causally linked events. Based on the cause-effect definition in Section 2, we provide them with the following criteria to guide their decision about whether an event A causes another event B:

1. a causal relation must be temporal, but a temporal relation is not always causal;

2. the action/state of A directly leads to the action/state of B i.e., there must be no intermediate events or if there is one, it should be extremely rare in real-world scenarios;

3. an event B would not occur if A did not occur.

Initially, the workers provide non-contextual annotations based solely on their commonsense understanding of the abstractions. A relation is deemed valid if the annotator can envision a plausible scenario in daily life where the situation occurs frequently, commonly, and is highly likely. If no such scenario comes to mind, the clusters are considered unrelated. In the subsequent step, we identify the highly disagreed pairs, where the three annotators each make distinct decisions regarding causality i.e., A causes B, B causes A, A and B are unrelated. For these pairs, workers are presented with contextual information from stories in GLUCOSE and asked to reconsider their decisions. The final determination of the relationship is made through majority voting.

C Clustering Algorithm

Our clustering algorithm, named PIVOT, is inspired by the pivoting algorithm proposed in Fukunaga (2019). The PIVOT algorithm first randomly selects a pair of cause-effect events and then, for each of them, find its most similar neighbors against a threshold of 70%. We repeat the process for the remaining event mentions, while excluding the previously assigned events. The initial results are passed to the following process to remove self-loop and bi-directions. We remove clusters with fewer than 10 samples and maximum pairwise similarity is less than 50%. Each cluster can now be considered a node in a graph and we use GLUCOSE to recover the causal relations among them to construct a temporary causal graph.

Ablation study. The main motivation behind PIVOT algorithm is to ensure the initial graph is mostly acyclic while avoiding any sub-optimality produced from post-processing. To validate whether PIVOT is most effective in ensuring causal consistency, we conduct an ablation study against popular clustering algorithms, including OPTICS (Ankerst et al., 1999), LOUVAIN (Blondel et al., 2008) and LEIDEN (Traag et al., 2019) algorithms, where LOUVAIN and LEIDEN were proposed for community detection problems. The criteria for selecting these clustering algorithms include: (1) scalability to medium-to-large-sized data, (2) ability to accommodate custom affinity matrix and (3) high cluster homogeneity score. Table 7 further reports the quality of the algorithms under analysis, which shows that our PIVOT algorithm yields the most desirable performance.

Notations. We use lower case letters (i.e., v) to denote single event, capital letters (i.e., V) for cluster of events, and blackboard bold letter (i.e., $\mathbb{V}$) for set of clusters. We let $\mathcal{D}$ denote the dataset of causal event mentions; $x \rightarrow y$ indicates event x is a cause of event y; $x \leftarrow y$ indicates event x is an effect of event y; $x \leftrightarrow y$ indicates x and y are causally related (either cause or effect). We also define the similarity between an event y and cluster V as the average of similarity between y and every event x in V

$$S_{yV} = \frac{1}{|V|} \sum_{x \in V} S_{xy},$$

where S_{xy} is the similarity score between two events according to Eq. (1).

Performance metrics. In the following, we describe the unsupervised performance metrics to assess clustering algorithms in Table 7. Given a set of clusters $\mathbb{C}$, let $\boldsymbol{A}$ be the matrix where $\boldsymbol{A}_{ij}$ is the number of events in cluster $C_i \in \mathbb{C}$ is the cause of any event in the cluster $C_j \in \mathbb{C}$. Recall that in this stage the causal relations between events are extracted from GLUCOSE dataset. A cluster A is said to cause another cluster B if at least one event mentions in cluster A causes any other event mentions in cluster B, according to the cause-effect definition in Section 2.

1. *Self-loop ratio*: Proportion of clusters in which the events are either cause or effect of each other.

$$\frac{1}{|\mathbb{C}|} \sum_{i=1}^{|\mathbb{C}|} \frac{\boldsymbol{A}_{ii}}{2|C_i|}.$$

2. *Bi-directional ratio:* Proportion of cluster pairs that are both cause and effect of one another.

$$\frac{2}{|\mathbb{C}|^2 - |\mathbb{C}|} \sum_{i=1}^{|\mathbb{C}|-1} \sum_{j=i+1}^{|\mathbb{C}|} \frac{\min(\boldsymbol{A}_{ij}, \boldsymbol{A}_{ji})}{\max(\boldsymbol{A}_{ij}, \boldsymbol{A}_{ji})}.$$

3. *Silhouette coefficient (Rousseeuw, 1987):* Measure of how similar an instance is to its own cluster (cohesion) compared to other clusters (separation). A high value indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters.

$$\frac{1}{|\mathcal{D}|}\sum_{x\in\mathcal{D}}\frac{a_x - b_x}{1-\min(a_x, b_x)},$$

where a_x is the mean similarity between event x and all other events in the same cluster; b_x is the mean similarity between event x and all other events in the next nearest cluster.

4. *Homogeneity score:* Average pairwise similarity of events in a cluster.

$$\frac{1}{|\mathbb{C}|}\sum_{i=1}^{|\mathbb{C}|}\frac{2}{|C_i|^2-|C_i|}\sum_{x,y\in C_i, x\neq y} S_{xy},$$

where S_{xy} is the similarity score between two events according to Eq. (1).

Table 8 reports the numerical results for the experiment on Abstract Event Identification in Section 4.1. For the supervised metrics, we refer readers to `scikit-learn`'s documentation for the precise formulations and implementations of *Adjusted Rand Index* (Steinley, 2004) and *Normalized Mutual Information* (Vinh et al., 2009).

D Statistical Causal Discovery

Background. The causal relations among n variables $X = [X_i]_{i=1}^n$ is characterized via a **structural causal model (SCM)** (Pearl, 2009) over the tuple $\langle U, X, f\rangle$ that, in its general form, consists of a sets of assignments

$$X_i := f_i\big(\mathrm{PA}_{X_i}, U_i\big), \quad i = 1, \cdots, n,$$

where U_i is an exogenous variable assumed to be mutually independent with variables $\{U_1, \cdots, U_n\}\backslash U_i$. The functions $f = [f_1, \cdots, f_n]$ define a joint distribution $P(X)$ over the endogenous variables X, given a joint distribution over exogenous variables $P(U_1, \cdots, U_n)$. Each SCM induces a causal graph $\mathbf{G}$, which is often assumed to be a DAG. A directed graph $\mathbf{G} = (\mathbf{V}, \mathbf{E})$ consists of a set of nodes $\mathbf{V}$ and an edge set $\mathrm{E} \subseteq \mathbf{V}^2$ of ordered pairs of nodes with $(v, v) \notin \mathbf{E}$ for any $v \in \mathbf{V}$ (one without self-loops).

For a pair of nodes i, j with $(i, j) \in \mathrm{E}$, there is an arrow pointing from i to j and we write $i \to j$. Two nodes i and j are adjacent if either $(i, j) \in \mathbf{E}$ or $(j, i) \in \mathbf{E}$. If there is an arrow from i to j then i is a parent of j and j is a child of i. Let PA_{X_i} denote the set of variables associated with parents of node i in $\mathbf{G}$. The graph $\mathbf{G}$ of an SCM is obtained by creating one vertex for each X_i and drawing directed edges from each parent $X_j \in \mathrm{PA}_{X_i}$ to X_i. We sometimes call the elements of PA_{X_i} the **direct causes** of X_i, and we call X_i a **direct effect** of each of its direct causes. Importantly, these functions are to be read as assignments rather than as mathematical equations, and they should be viewed as modelling physical mechanisms inducing or generating every X_i from variables PA_{X_i}.

Experiments. We here discuss how `ACCESS` is used to assess to what extent the statistical structure learning methods is applicable to recover causal relations among event abstractions. As illustrated in Figure 1, after extracting abstractions, one can build representations for abstract events in the original corpus and apply structure learning on top of such data for full graph discovery. A simple representation is the co-occurrence matrix size $(\#stories \times \#abstractions)$ where each entry takes a binary value indicating whether an abstraction has any of its mentions appearing in a story. This means each abstraction is now considered as a Bernoulli random variable and the task of causal discovery is to recover the underlying SCM where the structural functions are commonly non-convex.

Due to the limited scalability of existing statistical algorithms, we resort to learning sub-graphs by setting thresholds to select nodes that appear frequently while ensuring that the true graph is acyclic. Specifically, our selected sub-graphs are composed of edges where both nodes are adjacent to at least one other node, and each node corresponds to an abstraction whose occurrences exceed a certain frequency threshold. In our experiment, we set thresholds for document frequency within $\{25, 30, 35, 40, 45\}$, resulting in sub-graphs with $5, 7, 16, 19, 45$ nodes. The experiments are run on 5 CPU cores.

We experiment with popular constraint-based and score-based algorithms. We select those that are scalable and capable of capturing non-linear causal relationships without relying on specific model forms such as additive noise. In this paper, we report the results for the following algorithms: